Ex.No.8 - Decision Trees

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In [50]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.preprocessing import LabelEncoder
from sklearn.metrics import accuracy_score, confusion_matrix, ConfusionMatrixD
```

In [51]:
```
# Load Dataset
data = pd.read_csv(r"D:\VI\FOML\Datasets\Drug Prediction.csv")
data.head()
```

Out[51]:

	Age	Sex	BP	Cholesterol	Na_to_K	Drug
0	23	F	HIGH	HIGH	25.355	drugY
1	47	M	LOW	HIGH	13.093	drugC
2	47	M	LOW	HIGH	10.114	drugC
3	28	F	NORMAL	HIGH	7.798	drugX
4	61	F	LOW	HIGH	18.043	drugY

In [52]:
```
# Encode Categorical Columns
le_sex = LabelEncoder()
le_bp = LabelEncoder()
le_chol = LabelEncoder()
le_drug = LabelEncoder()

data['Sex'] = le_sex.fit_transform(data['Sex'])
data['BP'] = le_bp.fit_transform(data['BP'])
data['Cholesterol'] = le_chol.fit_transform(data['Cholesterol'])
data['Drug'] = le_drug.fit_transform(data['Drug'])

# Define Features and Target
X = data[['Age', 'Sex', 'BP', 'Cholesterol', 'Na_to_K']]
y = data['Drug']

X
```

Out[52]:

	Age	Sex	BP	Cholesterol	Na_to_K
0	23	0	0	0	25.355
1	47	1	1	0	13.093
2	47	1	1	0	10.114
3	28	0	2	0	7.798
4	61	0	1	0	18.043
...	...	...	...	...	...
195	56	0	1	0	11.567
196	16	1	1	0	12.006
197	52	1	2	0	9.894
198	23	1	2	1	14.020
199	40	0	1	1	11.349

200 rows × 5 columns

In [69]:

```
# Train Test Split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Train Decision Tree Model
model = DecisionTreeClassifier(
    random_state=42
)

model.fit(X_train, y_train)
model
```

Out[69]:

```
▾ DecisionTreeClassifier
DecisionTreeClassifier(random_state=42)
```

In [70]:

```
# Prediction and Accuracy
y_pred = model.predict(X_test)

print("Train Accuracy :", model.score(X_train, y_train))
print("Test Accuracy  :", model.score(X_test, y_test))
```

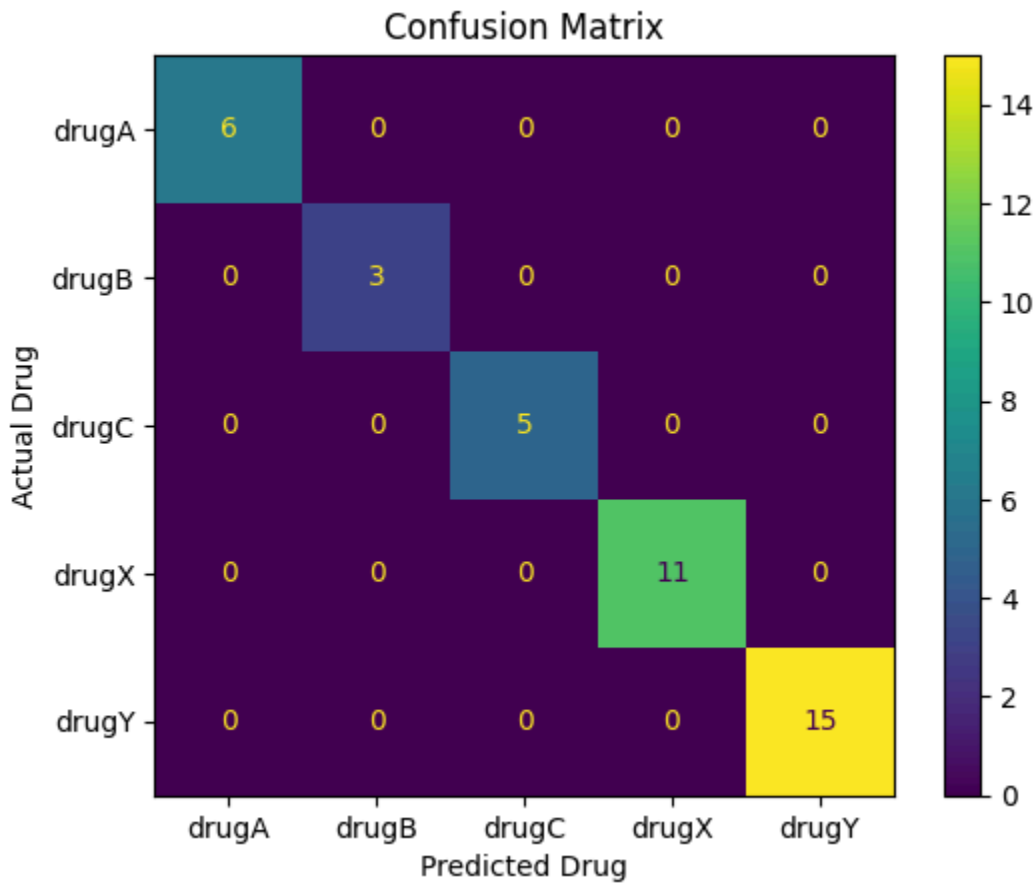
```
Train Accuracy : 1.0
Test Accuracy  : 1.0
```

In [79]:

```
# Confusion Matrix with Drug Names

cm = confusion_matrix(y_test, y_pred)
```

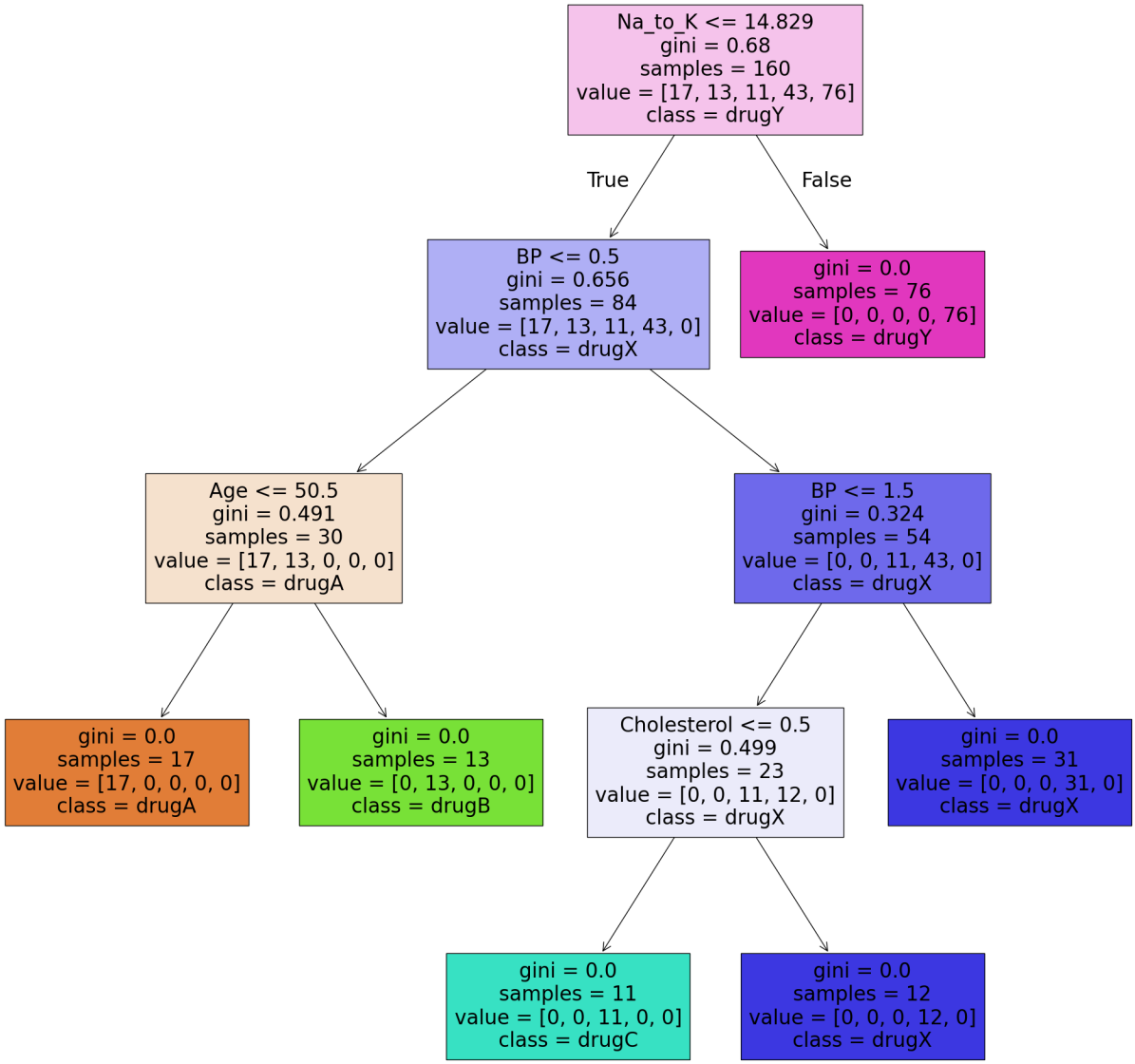
```
drug_names = le_drug.classes_

plt.figure(figsize=(8,6))

disp = ConfusionMatrixDisplay(
    confusion_matrix=cm,
    display_labels=drug_names
)

disp.plot()

plt.title("Confusion Matrix")
plt.xlabel("Predicted Drug")
plt.ylabel("Actual Drug")

plt.show()
```

```
<Figure size 800x600 with 0 Axes>
```

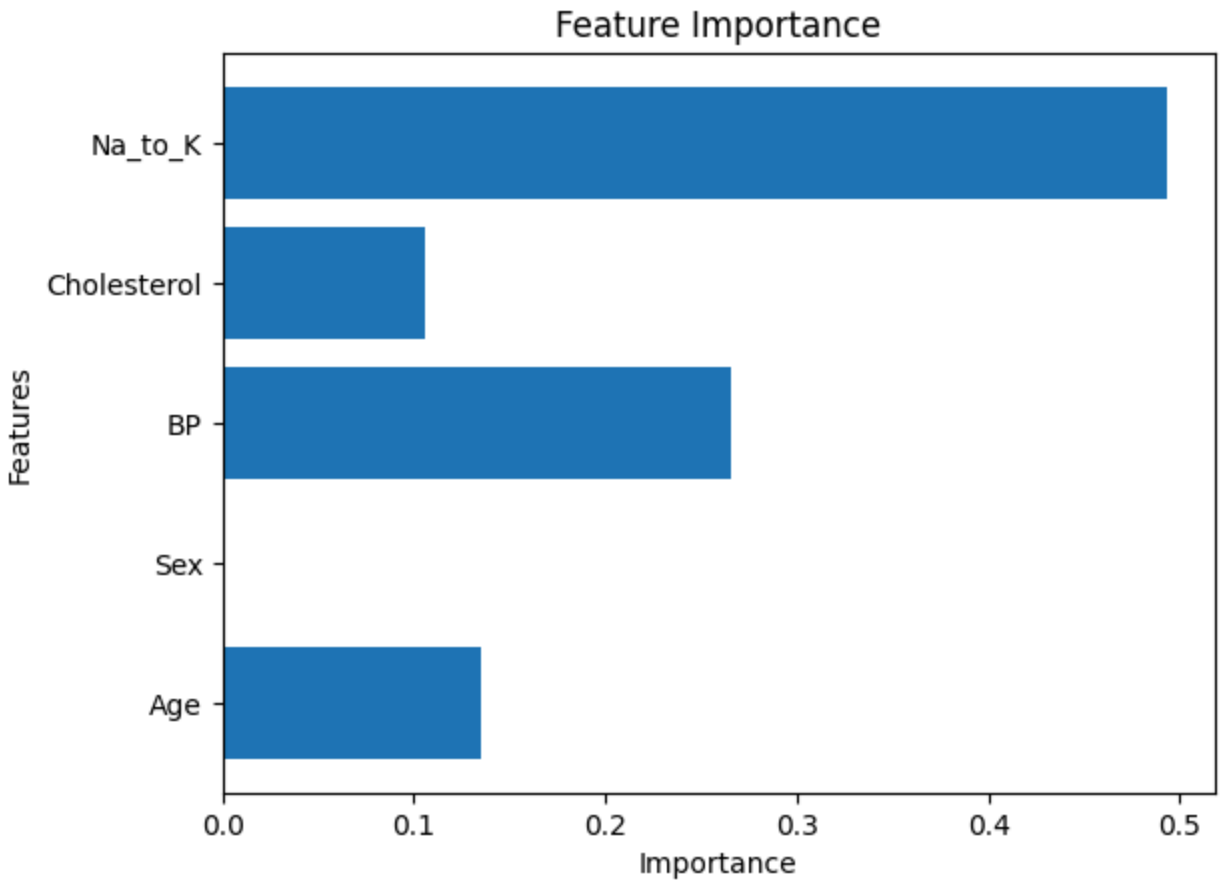


In [80]:
```
# Decision Tree Visualization
plt.figure(figsize=(25,25))

plot_tree(
    model,
    feature_names=X.columns,
    class_names=le_drug.classes_,
```

```
    filled=True
)

plt.savefig(r"D:\VI\FOML\Outputs\decision_tree.png", dpi=300, bbox_inches='tig
plt.show()
```



In [73]:
```
importance = model.feature_importances_

plt.figure()
plt.barh(X.columns, importance)
plt.xlabel("Importance")
plt.ylabel("Features")
plt.title("Feature Importance")
plt.show()
```



In [74]:
```
# Custom Data Prediction
# [Age, Sex, BP, Cholesterol, Na_to_K]

sample = [[47, 1, 0, 0, 14.5]]

prediction = model.predict(sample)

print("Predicted Drug :", le_drug.inverse_transform(prediction)[0])
```

```
Predicted Drug : drugA
```

```
C:\Users\Monish\AppData\Local\Programs\Python\Python312\Lib\site-packages\sklea
rn\base.py:493: UserWarning: X does not have valid feature names, but DecisionT
reeClassifier was fitted with feature names
  warnings.warn(
```

In [ ]: